Chapter 9 – Ancient Mathematical Discoveries

"We owe a lot to the Indians, who taught us how to count, without which no worthwhile scientific discovery could have been made."

(Albert Einstein)

Great building knowledge did not come without a solid understanding of geometry and mathematical concepts. By the 3rd century CE, Indians had begun using a decimal number system to express large values that may have been introduced by Chinese traders. The local origins of this system are unclear, but what's certain is that Indian mathematicians revolutionized the practice of using numbers from the third to fifth centuries. In fact, they created a set of numerals to use for the values 1 through 9 that would become the basis for counting and mathematics the world over. This set of numerals is credited to Aryabhata in the 5th or 6th century CE. We still use this Hindu-Arabic system today, regardless of regional language differences, making mathematics a truly universal language that spans nations.

Aryabhata also approximated the Pi ratio to within four decimal places. In doing so, he realized it was an irrational number, meaning that any calculation in which it was used could only be an approximation at best. Nevertheless, he saw the potential of the Pi approximation and used it to calculate the circumference of the earth to within 113 kilometers (70 miles) of the actual circumference. These calculations would be incredibly important for the future of mathematics and astronomy.

The mathematician also realized that he could arrange equations so that those with more than one missing value could be solved. This was particularly useful in working with right triangles, thanks to the sine function. Using sine—a specific ratio of sides of a right triangle that describe the precise measurement of acute angles—Aryabhata was able to disprove astronomers who proposed that the moon was farther from the earth than the sun. The proof was in the angles. If two individuals view the sun from different locations, they can gather enough information to create a triangular representation of their lookout positions and the sun. It can be shown this way (though not to scale):